BRAC UNIVERSITY

Inspiring Excellence

Assignment-01

(Boolean Simplification, CPOS, CSOP)

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Course: CSE260

Section: 17B

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Ans. to. the . ques . no - 1

$$F(W, X, Y, Z) = X\overline{Y} + \overline{X}YZ(X \oplus Z)$$

$$= X\overline{Y} + \overline{X}YZ(X\overline{Z} + \overline{X}Z)$$

$$= X\overline{Y} + \overline{X}YZ$$

$$= (X\overline{Y} + \overline{X})(X\overline{Y} + YZ)$$

$$= (\overline{X} + X)(\overline{X} + \overline{Y})(X\overline{Y} + Y)(X\overline{Y} + Z)$$

$$= (\overline{X} + \overline{Y})(X + Y)(\overline{Y} + Y)(X + Z)(\overline{Y} + Z)$$

$$= (\overline{X} + \overline{Y})(X + Y)(X + Z)(\overline{Y} + Z)$$

$$= (\overline{X} + \overline{Y} + Z\overline{Z} + W\overline{W})(X + Y + Z\overline{Z} + W\overline{W})(X + Z + Y\overline{Y} + W\overline{W})(\overline{Y} + Z + X\overline{X} + W\overline{W})$$

$$= (\overline{X} + \overline{Y} + W + Z)(\overline{X} + \overline{Y} + \overline{W} + Z)(\overline{X} + \overline{Y} + W + \overline{Z})(\overline{X} + \overline{Y} + \overline{W} + \overline{Z})$$

$$(X + Y + W + Z)(X + Y + \overline{W} + Z)(X + Y + W + \overline{Z})(X + Y + \overline{W} + \overline{Z})$$

$$(X + Z + Y + \overline{W})(X + Z + \overline{Y} + \overline{W})$$

$$(X + Z + Y + W)(X + Z + \overline{Y} + W)$$

$$(\overline{Y} + Z + X + \overline{W})(\overline{Y} + Z + \overline{X} + W)(\overline{Y} + Z + \overline{X} + \overline{W})$$

$$(\overline{Y} + Z + X + W)$$

$$= (\underset{0}{W}+\underset{1}{\overline{X}}+\underset{1}{\overline{Y}}+\underset{0}{Z})(\underset{1}{\overline{W}}+\underset{1}{\overline{X}}+\underset{1}{\overline{Y}}+\underset{0}{Z})(\underset{0}{W}+\underset{1}{\overline{X}}+\underset{1}{\overline{Y}}+\underset{1}{\overline{Z}})(\underset{1}{\overline{W}}+\underset{1}{\overline{X}}+\underset{1}{\overline{Y}}+\underset{1}{\overline{Z}})(\underset{0}{W}+\underset{0}{X}+\underset{0}{Y}+\underset{0}{Z})$$

$$\cdot(\underset{1}{\overline{W}}+\underset{0}{X}+\underset{0}{Y}+\underset{0}{Z})(\underset{0}{W}+\underset{0}{X}+\underset{0}{Y}+\underset{1}{\overline{Z}})(\underset{1}{\overline{W}}+\underset{0}{X}+\underset{0}{Y}+\underset{1}{\overline{Z}})(\underset{0}{W}+\underset{0}{X}+\underset{1}{\overline{Y}}+\underset{0}{Z})(\underset{1}{\overline{W}}+\underset{0}{X}+\underset{1}{\overline{Y}}+\underset{0}{Z})$$

$$= \Pi(6, 14, 7, 15, 0, 8, 1, 9, 2, 10)$$

$$= \Pi(0, 1, 2, 6, 7, 8, 9, 10, 14, 15)$$

Ans. to. the. ques. no - 2

$$F(P,Q,R,S) = ((PQ) \oplus (RS))(\bar{S} + \bar{P}QS)\bar{S}(Q \odot R) + \overline{((P+\bar{S}+Q)(\bar{S}+\overline{(Q+S)})R)}$$

$$= (PQ\overline{RS} + \overline{PQ}RS)(\bar{S} + \bar{P}QS)\bar{S}(QR + \bar{Q}\bar{R}) + (\overline{(P+\bar{S}+Q)} + \overline{(\bar{S}+\overline{(Q+S)})} + \bar{R})$$

$$= (PQ\overline{RS}\bar{S} + \overline{PQ}RS\overline{PQ})(\bar{S}QR + \bar{S}\bar{Q}\bar{R}) + (\bar{P}SQ + (S(Q+S)) + \bar{R})$$

$$= PQ\overline{RS}\bar{S}R + (\bar{P}SQ + QS + S + \bar{R})$$

$$= PQR\bar{S}(\bar{R}+\bar{S}) + (\bar{P}SQ + S + \bar{R})$$

$$= PQR\bar{S} + S + \bar{R}$$

$$= (S+\bar{S})(S+PQR) + \bar{R}$$

$$= S + PQR + \bar{R}$$

$$= (R+\bar{R})(\bar{R}+PQ) + S$$

$$= \bar{R} + PQ + S$$

ASSIGNMENT 1:

1. Convert the following function to canonical POS using distributive law and mention the maxterms: [2.5]

   $$F(W,X,Y,Z) = XY'+X'YZ(X \oplus Z)$$

   NB: You cannot use truth table or K-map to solve this question, you also cannot do canonical SOP first and then convert to canonical POS later.

2. Simplify the following function to the minimum number of literals using boolean algebraic formulas: [2.5]

   $$F(P,Q,R,S) = ((PQ) \oplus (RS))\ (S'+P'QS)\ S'(Q \odot R) + (\ (P+S'+Q)(S'+(Q+S)')\ R)'$$

SUBMISSION LINK: https://forms.gle/qsUBKRgrMXBPSxdYA

DEADLINE: 8 July, 2026 at 11 59 PM

In case of late Submissions, 10% marks will be deducted for every 12 hours of delay.